

Silver Plating + Faraday's Law

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Count the atoms you deposited

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 3 — Silver Plating and Faraday's Law

Hook: Work out how many silver atoms you deposited, and how thick the film is, without weighing anything.

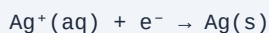
Learning outcomes

- Review galvanic vs electrolytic ideas from Sessions 1–2
- Distinguish a **reactant salt** (AgNO_3) from a **supporting electrolyte** (KNO_3) in the same beaker
- Plate silver onto copper at a **controlled current density**, and say why the value matters
- Use $Q = It$, $n = Q/F$, and the mass/thickness/atom-count relations
- Explain and apply **live entry**, and why copper spontaneously plates itself with silver without it
- Complete a Faraday prediction **before** plating, then test it against measured current, a control coin, and the pH

Session plan (90 min)

Time	Activity	Reference
0–8 min	Review Days 1–2 + hook	lecture.md §1, §3.1
8–18 min	Ag^+ reduction, two salts, Faraday pipeline, current density, live entry	lecture.md §3
18–26 min	Quiz check	lecture.md §4
26–42 min	Interactive pre-lab calculation (required gate)	lecture.md §5
42–52 min	Estimate area; clean; wire with resistor ; rehearse live entry	experiment.md — Parts A, B
52–70 min	Timed plating run + no-power control coin	experiment.md — Part C
70–80 min	Recalculate with real I_{avg} ; compare with prediction	experiment.md — Part D
80–86 min	Plated vs control; check pH against the charge	experiment.md — Part E
86–90 min	Debrief efficiency; bridge to Session 4	lecture.md §7

Key equations



$$Q = I \times t$$

$$n_{\text{e}} = Q / F \quad F = 96,485 \text{ C/mol}$$

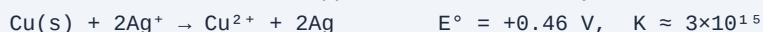
$$n_{\text{Ag}} = n_{\text{e}} \quad (z = 1)$$

$$m_{\text{Ag}} = n_{\text{Ag}} \times 107.87 \text{ g/mol}$$

$$d = m_{\text{Ag}} / (10.49 \times A)$$

$$N = n_{\text{Ag}} \times 6.022 \times 10^{23}$$

And the reaction that happens with no battery at all:



The bath

Component	Concentration	Job
AgNO ₃	0.05 M	The reactant — becomes the coating
KNO ₃	0.2 M	Supporting electrolyte — carries current, is not consumed

Distilled water only. Target current density **0.5–1.0 mA/cm²**, about **2 mA** on a typical coin, set by a **1 kΩ series resistor**.

Status

- ☐ KNO₃ chloride-tested against AgNO₃
- ☐ 0.05 M AgNO₃ / 0.2 M KNO₃ prepared with distilled water and labeled
- ☐ 1 kΩ resistors staged, one per station
- ☐ Two matched coins per group, one for the control
- ☐ Pilot run completed; live entry rehearsed
- ☐ Pre-lab worksheets printed (lecture.md §5)
- ☐ pH paper or meter checked against a buffer
- ☐ Silver waste container labeled, with a copper coil in it for recovery

Session 3 — Preparation

The three things that decide whether this session works

1. **The bath is 0.05 M AgNO_3 in 0.2 M KNO_3** , not dilute silver nitrate alone. The silver concentration sets the maximum current the cell can sustain; the nitrate makes the solution conduct.
 2. **A 1 k Ω resistor goes in series at every station.** It sets the current, so all benches match and nobody has to chase a target by fiddling with electrode depth.
 3. **Live entry.** The object is powered before it touches the liquid and stays powered until it leaves. Otherwise copper displaces silver directly and the coating will not adhere.
-

Before class (instructor)

Solution

- [] **Chloride-check the KNO_3** against AgNO_3 on a watch glass — clear, not milky
- [] Mix the bath with **distilled water**: dissolve KNO_3 first, then AgNO_3
- [] Enough for all groups at 30–40 mL each, plus 20%
- [] Label with concentration and date; store dark
- [] Confirm the silver waste protocol with the facility

Pilot — do this properly, it takes fifteen minutes

- [] Plate one object end to end and record current, time, and appearance
- [] Confirm the **1 k Ω resistor** gives about **2 mA** with your 3 V packs
- [] Confirm the current stays flat across the run
- [] Practise **live entry** yourself before teaching it
- [] Run a **control coin with no power** for the same time and keep it — it is your demonstration piece
- [] Measure start and end pH and check them against the charge
- [] Photograph both coins side by side

Hardware

- [] 1 k Ω resistors, one per station plus spares
- [] Clean graphite anodes
- [] Multimeter fuses checked — a blown mA fuse reads zero and looks like "no reaction"
- [] Two matched, cleaned coins per group
- [] Print pre-lab worksheets (lecture.md §5) and station cards

AgNO_3 handling

- Gloves when preparing; ventilated area
- Label everything; store away from organics and light
- Have the ammonia warning ready to say out loud — see the safety section in experiment.md

Day-of setup

- [] Pre-measured bath volumes at each station, to reduce spills
 - [] Resistors already inserted in the leads, so nobody forgets
 - [] Goggles enforced at the door
 - [] Silver waste container visible and labeled
 - [] Worked example on the board before students start
 - [] Control-coin dishes at each bench
 - [] Last year's dendrite photo displayed
-

Common failure modes

Issue	Prevention
Coating rubs off	Object entered the bath unpowered. Station a TA at each bench for the first entry
Grey fluffy deposit	Current density too high. Check the resistor is actually in circuit and the area estimate is right
Bath cloudy from the start	Chloride — tap water or contaminated nitrate. Chloride-check beforehand
Every bench a different current	Resistor missing, or someone set a voltage instead
Meter reads zero	Blown mA fuse, or not in series
Calculations stall	Groups that skipped the \$5 pre-lab gate. Enforce it
pH reading drifts or looks wrong	Silver attacks glass electrodes with KCl references. Calibrate immediately before, rinse well after, or use paper
Someone proposes adding ammonia	Pre-empt it in the safety brief. Ammoniacal silver forms explosive solids on standing

Timing note

Calculations take longer than expected. Have the worked example on the board as a check, and consider a spreadsheet template for groups that finish the arithmetic early — they can recompute for a different current and see how thickness scales.

Open questions / notes for next year

- Bath concentration used, and whether 0.05 M was necessary or 0.025 M would have done:
- Resistor value and the current it actually gave:
- Measured current density:
- Appearance — even? matte or bright? Did burnishing help?
- Did the plated coin survive a rub, and did the control fail one?

- Predicted vs measured pH:
- Volume of AgNO_3 consumed across the class:
- Was live entry followed reliably, or did groups need reminding?

Session 3 — Materials

Per group

- ☐ 30–40 mL of plating bath: 0.05 M AgNO_3 / 0.2 M KNO_3 (instructor prepares)
- ☐ Distilled water for rinsing
- ☐ **Two** objects to plate — copper coins or washers, matched. One is plated, one is the **no-power control**
- ☐ **Graphite** electrode (anode — inert, does not replenish Ag^+)
- ☐ 3 V battery pack
- ☐ **1 k Ω resistor** — required, one per station
- ☐ Multimeter (DC mA)
- ☐ Alligator clips
- ☐ Stopwatch
- ☐ Small beaker or jar, plus a second small dish for the control
- ☐ Ruler or calipers (area estimation)
- ☐ Sandpaper or steel wool
- ☐ Gloves, goggles, forceps
- ☐ Soft cloth for burnishing
- ☐ pH paper or a meter (start and end readings)

Solution preparation — instructor

Target: 0.05 M AgNO_3 in 0.2 M KNO_3 . Distilled water only.

Batch	AgNO_3	KNO_3	Or NaNO_3 instead
250 mL	2.12 g	5.05 g	4.25 g
500 mL	4.25 g	10.11 g	8.50 g

Half-strength silver option (if AgNO_3 is scarce): 0.025 M AgNO_3 — 1.06 g / 250 mL, 2.12 g / 500 mL — with the same KNO_3 , and halve the target current to about 1 mA.

Dissolve the nitrate salt first, then the AgNO_3 . Label 0.05 M AgNO_3 / 0.2 M KNO_3 — gloves — silver waste only with the date. Store dark.

Chloride check on your KNO_3 — before mixing a full batch

Combine a few drops of dissolved KNO_3 with a few drops of AgNO_3 solution on a watch glass. **Clear** = good. **Milky white** = chloride contamination; find a purer grade. Two minutes here saves a ruined batch.

Instructor-only

- ☐ Solid AgNO_3 — students never handle it
- ☐ Dedicated **silver waste** container, with a coil of bare copper wire in it for recovery

- [] Spare gloves (staining)
- [] Printed pre-lab worksheets — lecture.md §5
- [] Last year's photo of the dendritic overnight failure, for \$3.8

Optional

- [] Magnetic stirrer or stir sticks — raises the current ceiling and improves the deposit
- [] Balance (see the note below before promising an efficiency measurement)
- [] Silver anode — sterling wire or a bullion round. Eliminates pH drift and bath depletion entirely; a reusable purchase worth considering

A note on the balance

The predicted deposit is about **2 mg**. A ± 0.001 g balance resolves that to roughly $\pm 50\%$, so a mass-based efficiency check is not really achievable here. Either use a ± 0.1 mg balance or skip it — and tell students *why* it is being skipped. Session 4 measures a gas volume instead, which is where the efficiency question finally gets a real answer.

Waste

- [] All silver-bearing liquid to the labeled container — never the drain
- [] Copper coil left in the waste bottle strips the silver back out as metal ($K \approx 3 \times 10^{15}$); filter and keep it

Pilot notes — fill in for next year

Parameter	Value
Bath concentration used	
Resistor used	_____ Ω
Measured current	_____ mA
Current density	_____ mA/cm ²
Plating time	_____ s
Appearance: even? matte or bright?	
Did it survive a rub?	
Control coin appearance	
Starting / final pH	___/___
Predicted pH	_____

Session 3 — Experiment: Silver Plating + Calculations

Instructors: bath design, the displacement problem, limiting current, live entry — instructor.md.

Session arc

1. **Part A** — Estimate the plated area
2. **Part B** — Build the cell: series resistor sets the current, live entry protects the copper
3. **Part C** — Timed plating run, with a **no-power control coin** running alongside
4. **Part D** — Faraday calculations, compared against the prediction
5. **Part E** — Compare plated vs control; measure the pH change and check it against the charge

The bath — 0.05 M AgNO₃ in 0.2 M KNO₃

Two dissolved things, doing two different jobs. Students should be able to say which is which.

Component	Job	Consumed?
AgNO ₃ , 0.05 M	Supplies the Ag ⁺ that becomes the coating	Yes — it is the reactant
KNO ₃ , 0.2 M	Supporting electrolyte: carries current through the solution	No — spectator, like Session 4's sulfate

Recipe

Distilled water only. Tap chloride precipitates AgCl and ruins the bath.

Batch	AgNO ₃ (0.05 M)	KNO ₃ (0.2 M)
250 mL	2.12 g	5.05 g
500 mL	4.25 g	10.11 g

Sodium nitrate works identically if you have it instead: 4.25 g / 250 mL, 8.50 g / 500 mL.

Half-strength option if silver is tight: 0.025 M AgNO₃ (1.06 g / 250 mL) with the same KNO₃, and halve the target current to about 1 mA.

Mixing order: dissolve the KNO₃ first, then add the AgNO₃ and stir until clear. Label 0.05 M AgNO₃ / 0.2 M KNO₃ — gloves — silver waste only with the date.

Chloride check on the KNO₃ — 30 seconds, do it before mixing a full batch

Any chloride contamination in your nitrate salt will fog the bath with AgCl. Put a few drops of your dissolved KNO₃ into a few drops of AgNO₃ solution on a watch glass.

- **Stays clear** → good, proceed.
- **Goes milky white** → your nitrate is contaminated. Find a purer grade; do not use it.

Volume per group

30–40 mL — enough to cover the object with the electrodes 2–3 cm apart.

Part A — Estimate the plated area (42–47 min)

You cannot calculate a thickness without an area, and area is where most groups go wrong.

Coin, one face: measure diameter d in cm $\rightarrow A = \pi(d/2)^2$. **Both faces plate:** $A = 2\pi(d/2)^2$. **Washer, one side:** $A \approx \pi(R^2_{\text{outer}} - R^2_{\text{inner}})$. **Irregular:** approximate the immersed part as length $\times$ width and write down that you approximated.

Dimension	Value (cm)	Formula used	Area A (cm ²)

Tip: a typical coin face is 2–4 cm². If you calculated 12 cm², you used the diameter where the radius belongs.

Part B — Build the cell (47–52 min)

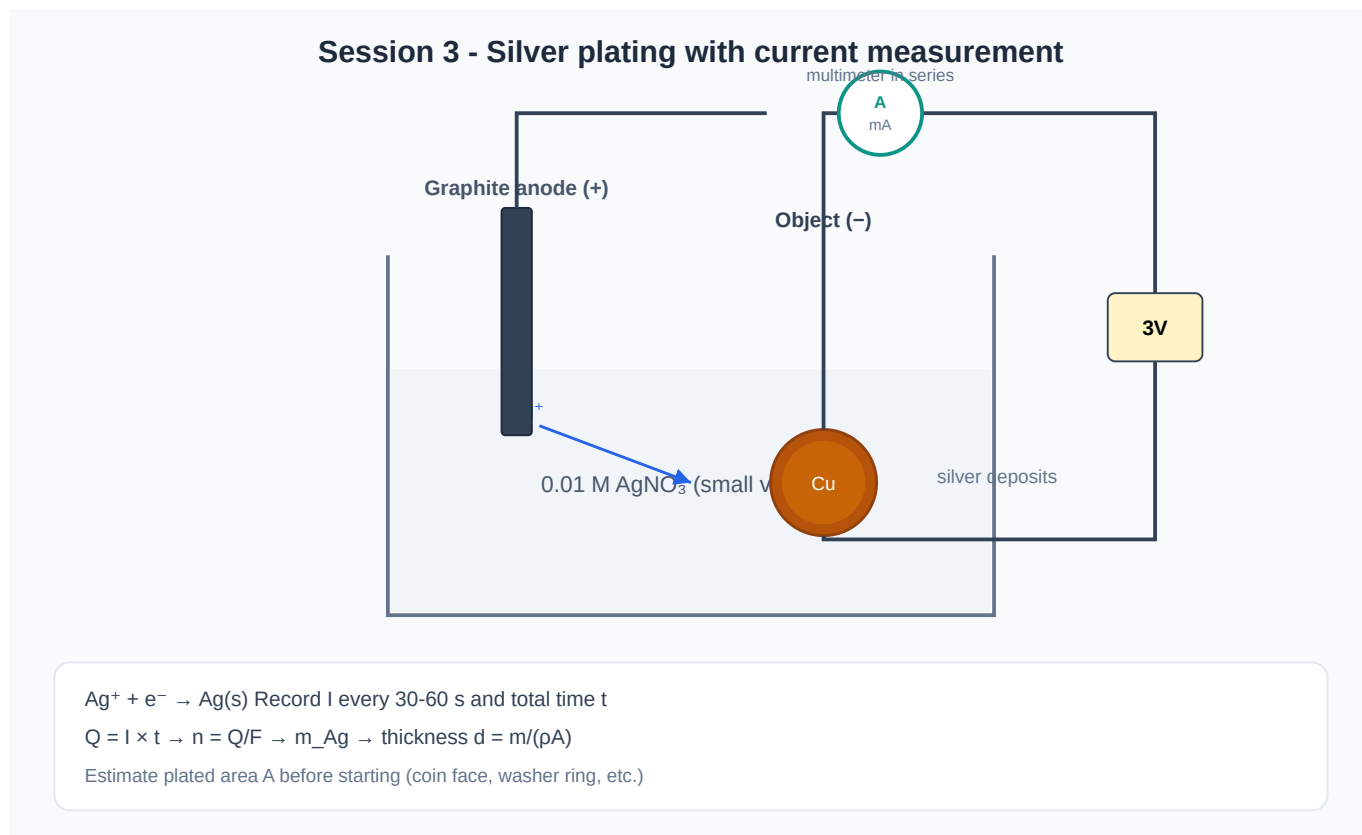


Figure 1 — Object on (–) as the cathode, graphite anode on (+), ammeter in series, and a resistor that fixes the current.

The circuit

Part	Connects to
Object to plate	Battery (–) — this makes it the cathode

Part	Connects to
Graphite anode	Battery (+)
1 k Ω resistor	In series, on either lead — required, not optional
Multimeter on DC mA	In series

Why the resistor is not optional

Last year this cell was run by setting a voltage and hoping. It does not work, because the current then depends on how deep the electrodes are, how far apart they are, and how the solution is behaving — so every group gets a different current and half of them get a bad deposit.

With 0.2 M KNO₃ the solution barely resists current at all, so a **1 k Ω resistor in series completely dominates the circuit and sets the current for you:**

$$I \approx (3 \text{ V} - \text{about } 1 \text{ V across the cell}) / 1000 \, \Omega \approx 2 \text{ mA}$$

Change the resistor to change the current, and nothing else matters much. That is what makes this experiment reproducible across eight benches.

Resistor (3 V pack)	Approximate current
2.2 k Ω	~1 mA
1 kΩ	~2 mA — use this
470 Ω	~4 mA

Target current density — the number that decides whether it works

Aim for **0.5–1.0 mA/cm²**, which for a typical 3 cm² coin means about **2–3 mA**.

Go much above that and you exceed the rate at which Ag⁺ can diffuse to the surface. Then the deposit stops being a coating and starts being a **dendrite** — a grey fluffy tree that grows off the tips and falls off. Gentle stirring raises the ceiling, so stir if you can.

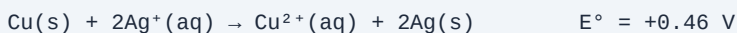
Clean the object

1. Sand or steel-wool until bright metal shows all over.
2. Rinse with distilled water, pat dry.
3. **Do not touch the surface to be plated.** Handle by the edge with forceps. Finger oil ruins adhesion.

Live entry — the step that decides whether the coating sticks

Connect the circuit and confirm current is flowing BEFORE the object touches the solution.

Here is why. Copper spontaneously reacts with silver ions the instant they meet, with no power at all:



That reaction deposits a loose, powdery, badly stuck layer of silver — and anything you plate on top of it flakes off along with it. Powering the object first makes it a cathode from the moment it gets wet, which stops the copper dissolving and stops the loose layer forming.

The sequence, in order:

1. Wire everything up with the object held **above** the solution.
2. Switch on. Confirm the ammeter reads a small current (it will read near zero until the circuit closes through the liquid).
3. **Lower the object in while powered.** Start the stopwatch as it goes under.
4. At the end: **lift the object out while still powered**, then switch off.

Never let the object sit in the bath unpowered. Not before, not after, not "just for a second."

Setup record

Item	Value
Object	
Area A (cm ²)	
Bath	0.05 M AgNO ₃ / 0.2 M KNO ₃
Solution volume (mL)	
Starting pH	
Anode	Graphite
Series resistor (Ω)	
Supply voltage	
Target time (s)	900 s (15 min)
Initial current (mA)	
Current density (mA/cm ²)	I/A =

Part C — Timed run, plus the control (52–70 min)

Set the control going at the same moment

Take a **second, identically cleaned coin**. Drop it into a small dish of the same bath with **no wires attached at all**, at the same time you start plating. Leave it the full 15 minutes.

This coin is the experiment's most valuable single piece of data. It shows you exactly what the copper does on its own, so that at the end you can tell the difference between silver you *plated* and silver that simply *arrived*.

The plating run

1. Start the stopwatch as the powered object enters the solution.
2. Log the current every 60 s.

3. Stir gently every couple of minutes, or leave a slow stirrer running.
4. At 900 s, lift the object out **while powered**, then switch off.
5. Rinse with distilled water, pat dry. Do not scrape it.
6. **Burnish** lightly with a soft cloth or a soft brush. A matte deposit comes up noticeably brighter — this is normal practice, not cheating.

Current log

Time (s)	I (mA)	I (A) = mA ÷ 1000	Notes
0			
180			
360			
540			
720			
900			

Average current: $I_{\text{avg}} = __ \text{ mA} = ____ \text{ A}$

With the resistor in place the current should be nearly flat. **A steady current is itself a result** — last year's cell, without a resistor and without supporting electrolyte, drifted all over the place.

Part D — Faraday calculations (70–80 min)

Use I_{avg} in amperes and t in seconds.

Quantity	Value
Faraday constant F	96,485 C/mol
Molar mass Ag	107.87 g/mol
Density $\rho(\text{Ag})$	10.49 g/cm ³
Electrons per Ag^+	$z = 1$

Step	Formula	Your calculation	Result
1. Charge	$Q = I_{\text{avg}} \times t$		_____ C
2. Moles e^-	$n_e = Q / 96,485$		_____ mol
3. Moles Ag	$n_{\text{Ag}} = n_e$ (since $z = 1$)		_____ mol
4. Mass Ag	$m = n_{\text{Ag}} \times 107.87$		_____ g
5. Volume Ag	$V = m / 10.49$		_____ cm ³
6. Thickness	$d = V / A$		_____ cm

Step	Formula	Your calculation	Result
7. In micrometres	$d \times 10,000$		_____ μm
8. Atoms	$n_{\text{Ag}} \times 6.022 \times 10^{23}$		_____ atoms

Worked check (instructor board version)

$I = 0.0020 \text{ A}$, $t = 900 \text{ s}$, $A = 3.14 \text{ cm}^2$:

$Q = 1.80 \text{ C} \rightarrow n_{\text{e}} = 1.87 \times 10^{-5} \text{ mol} \rightarrow m = 2.01 \text{ mg} \rightarrow V = 1.92 \times 10^{-4} \text{ cm}^3 \rightarrow d = 0.61 \mu\text{m} \rightarrow 1.1 \times 10^{19} \text{ atoms}$

Two milligrams of silver, six tenths of a micrometre thick, eleven billion billion atoms. All three of those numbers describe the same object.

Part E — Compare with reality (80–86 min)

E1 — Your coin versus the control coin

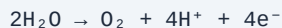
Put them side by side.

	Plated coin (powered)	Control coin (no power)
Colour		
Even or patchy?		
Shiny or matte/grey?		
Does it survive a firm rub with a cloth?		

The control will have picked up silver too — that is the displacement reaction — but it should be **duller, greyer, patchier, and it should rub off**. The powered coin should be more even and should stay put. If they look the same, your live entry did not work, and the honest conclusion is that most of your coating arrived chemically rather than electrically.

E2 — Check the pH against the charge

The graphite anode is splitting water and releasing protons:



So the moles of H^+ released should equal the moles of electrons you already calculated in step 2.

Quantity	Value
Moles of e^- (from step 2)	_____ mol
Bath volume in litres	_____ L
Predicted $[\text{H}^+]$ added = mol / L	_____ M
Predicted pH = $-\log[\text{H}^+]$	_____
Measured final pH	_____

For the worked example above in 30 mL, this predicts **pH $\approx$ 3.2**, down from about 7.

If your measurement lands near the prediction, you have just confirmed that essentially every electron did the job you assigned it. If the measured pH is much *higher* than predicted, then something at the cathode was consuming protons instead of depositing silver — which is exactly what happens when a bath is run too hard or for too long and runs out of Ag^+ .

Discussion prompts

1. Can you *see* a layer 0.6 μm thick? What does the calculation actually predict you will observe?
2. Did your current stay flat? What would have made it drift?
3. Your coin and the control coin both gained silver. Which process do you trust, and why?
4. What would happen if you left this running overnight?

Troubleshooting

Problem	Cause	Fix
Grey fluffy growth instead of a coating	Current density too high — you are past the diffusion limit	Bigger resistor; stir; check your area estimate
Coating rubs straight off	Displacement layer underneath — object entered unpowered	Re-clean and repeat with live entry
Deposit black rather than silvery	Very fine silver, from too much current	Lower the current; burnish afterwards
Bath cloudy white from the start	Chloride — tap water, or contaminated KNO_3	Remake with distilled water and tested nitrate
Bath turns faintly blue over the day	Cu^{2+} from displacement	Expected; a reason not to leave objects sitting in the bath
Current much lower than expected	Bad clip, meter fuse, electrodes not both immersed	Check the fuse first
Current drifting downward	Poor contact, or bath running low on silver	Check clips; at 0.05 M depletion is under 2% per run, so suspect the contact
Calculated thickness absurd	mA not converted to A, or area wrong	Recheck both; 0.1–2 μm is the sane window

Safety

- **Gloves and goggles mandatory.** AgNO_3 stains skin and clothing dark brown-black and the stain is effectively permanent — it is metallic silver forming in your skin
- Instructor prepares and dilutes all silver solutions; students never handle solid AgNO_3
- Small volumes only, 30–40 mL per group
- Keep the bath out of direct light; silver nitrate is light-sensitive

- **Never add ammonia to silver nitrate.** Ammoniacal silver solutions form explosive silver nitride on standing. Online "silver mirror" recipes use exactly this — do not follow them
- All silver-bearing liquid goes to the labeled **silver waste** container, never down the drain

Silver recovery — worth doing, and it teaches something

Drop a coil of bare copper wire into the silver waste bottle and leave it. The same displacement reaction that spoiled last year's overnight run will strip essentially every silver ion out of the waste and leave it as grey metallic fluff on the wire, which you can filter and keep. The equilibrium constant is about 3×10^{15} , so it goes to completion.

The reaction that was the problem in the cell is the standard industrial method for recovering silver from spent solution. That is worth pointing out.

Experiment status

- ☐ KNO_3 chloride-tested against AgNO_3 (stayed clear)
- ☐ 0.05 M AgNO_3 / 0.2 M KNO_3 mixed with distilled water and labeled
- ☐ 1 k Ω resistors, one per station
- ☐ Pilot run done: current __ **mA**, **appearance** ____
- ☐ Live entry rehearsed by every TA
- ☐ Second cleaned coin per group, for the no-power control
- ☐ pH meter or paper checked against a known buffer
- ☐ Silver waste container labeled; copper coil in it for recovery

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

The three rules that matter most

1. **Salt belongs in exactly one place this week.** NaCl is used only for the Session 5 corrosion jars, where nothing is driven hard enough to make chlorine. It is **forbidden** in every electrolysis bath and in the derusting cells, where chloride is oxidized at the anode to **chlorine gas**. This is a competing half-reaction, not an arbitrary rule.
2. **Any cell left running unattended must be open and ventilated.** Electrolysis and derusting cells evolve hydrogen continuously. An overnight derusting cell can generate a couple of litres of it. Open jars, ventilated room, no ignition sources, nothing sealed. Ever.
3. **Disconnect power before touching anything wet.** Every time, as a ritual, especially at the Session 5 reveal.

General rules

1. **No eating experimental materials.** Fruit, potatoes and liquids used in experiments are no longer food, and this includes the ones that started out edible.
2. **Ammeters go in series.** A meter on a current range clipped across a supply is a short circuit and will blow its fuse at best.
3. **Avoid short circuits generally.** Batteries heat up fast when shorted.
4. **Collect metal-salt waste.** Copper and silver solutions are not poured down the drain.
5. **Water choice:** tap water is fine for most baths. **Distilled is required for the Session 3 silver bath**, so chloride does not precipitate silver, and is worth having for the first rung of the Session 4 conductivity ladder.
6. **Goggles for all wet work.** Gloves additionally for silver nitrate.
7. **Rusty hardware is sharp** and often outdoors-dirty. Forceps or gloves; no bare-handed scrubbing.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails and wires	Supervise handling; wash hands after handling metals
2	Copper sulfate solution	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves and goggles; stains skin and clothing near-permanently; small volumes (30–40 mL); collect all waste; no ammonia, ever
4	Gas evolution; hydrogen ignition	No NaCl; ammeter in series; U-tube side arms open; balloon and flame instructor only
5	Corrosion demos; overnight derusting	Washing soda only, never NaCl; disconnect before handling; jars open and ventilated overnight

Silver nitrate and ammonia — Session 3

Never add ammonia, or any ammonia-containing cleaner, to a silver nitrate solution. Ammoniacal silver solutions (Tollens' reagent) deposit **silver nitride and related compounds** on standing, which are friction- and shock-sensitive explosives. The hazard is not the fresh solution — it is any residue left in a bottle, on a stirrer, or around a cap.

This matters because the "silver mirror" recipes students will find online use exactly this chemistry. Say so out loud during the Session 3 safety brief, before someone tries it at home.

It also does not even work for our purpose: ammonia complexes copper about as strongly as it complexes silver, so it does nothing to suppress the displacement reaction we are trying to control.

Hydrogen — Session 4, instructor only

The balloon demo is optional. The chemistry is fully demonstrated by the splint tests without any of this risk, so skip the balloon entirely if the venue forbids open flame or if you are at all unsure.

If you do run it:

- Feed **cathode (H₂) gas only** into the balloon. The anode vents separately to the room.
- **Never a deliberate H₂/O₂ mixture.** Pure hydrogen in air *deflagrates* — it burns at the surface and gives a bark. A stoichiometric mixture **detonates**, which is a different event entirely and not a classroom demonstration.
- **Never a rigid or sealed container.** A balloon tears; glass becomes shrapnel.
- Modest inflation only.
- Tie off, then carry it **well away** from the electrolysis cell, which is still making hydrogen.
- Long lighter at arm's length; area cleared; extinguisher present and its location known.
- Warn the room. The bang is loud, sudden, and some students find it distressing.

Splint tests — the low-risk alternative

Gas	Test	Result
H ₂	Lit splint at the mouth of the tube	Squeaky pop
O ₂	Glowing (not flaming) splint inserted	Ember relights

A few millilitres of gas in a small test tube is plenty. Keep hydrogen tubes **mouth-down** until the moment of testing. TA-supervised; students may hold the tube, an adult holds the flame.

Waste

Waste	Disposal
Copper sulfate solution	Labeled waste container; not down the drain
Silver nitrate solution and rinses	Labeled waste container; not down the drain. Drop a coil of bare copper wire in — it strips the silver back out as metal for recovery
Sodium sulfate electrolyte	Drain is acceptable; dilute well

Waste	Disposal
Washing soda derusting sludge	Let solids settle, bag the sludge, dilute the liquid to drain
Saltwater corrosion jars	Drain; dispose of rusty metal in solid waste